Hajime Mihara

R&D Engineer at Sony

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Summary

I am an R&D Engineer at Sony and hold an M.E. in Information Science from the Nara Institute of Science and Technology (NAIST).

My research interests lie in computer vision, computational photography, and computational imaging. Throughout my career, I have consistently worked at the interface between emerging image sensors, such as polarization, time-of-flight (ToF), and near-infrared sensors, and advanced signal processing, including deep learning and Vision Foundation Models.

I have published at leading international conferences such as CVPR and ICCP and actively patent my research outcomes, with an emphasis on technologies that can be translated into products and real-world applications. I also collaborate with business units developing media-production technologies, helping turn research innovations into practical systems, technology demonstrations, and public exhibitions.

Work Experience

R&D Engineer | *Computer Vision, Image Processing, Computational Photography* — Apr. 2016 – Present

Sony — Tokyo, Japan

- **Oct. 2022 – Present:** **R&D Center, Sony Semiconductor Solutions Corporation**
- **Apr. 2026 – Present:** **Imaging System Technology Div., Sony Corporation (concurrent)**
- Apr. 2021 – Sept. 2022: R&D Center, Sony Group Corporation (trade name change)
- Apr. 2016 – Mar. 2021: R&D Center, Sony Corporation

Education

Master of Engineering | *Computer Vision, Computational Photography* — Apr. 2014 – Mar. 2016

Nara Institute of Science and Technology (NAIST) — Nara, Japan

- Thesis: "Graph-cut based Semi-automatic Region Selection for 4D Light-field Editing"
- Advisers: Prof. Yasuhiro Mukaigawa, Assoc. Prof. Takuya Funatomi

Bachelor of Engineering | *Electrical and Electronics Engineering* — Apr. 2012 – Mar. 2014

Advanced Course, National Institute of Technology, Asahikawa College — Hokkaido, Japan

- Supervisor: Prof. Kohzoh Ohshima

Associate Degree | *Electrical and Electronics Engineering* — Apr. 2007 – Mar. 2012

National Institute of Technology, Asahikawa College — Hokkaido, Japan

Publications

Google Scholar: 128 citations, h-index 5, i10-index 4 (as of 2026).

1. C. Li, T. Ono, T. Uemori, S. Nitta, **H. Mihara**, A. Gatto, H. Nagahara, Y. Moriuchi, "NeISF++: Neural Incident Stokes Field for Polarized Inverse Rendering of Conductors and Dielectrics," IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 26493–26503, 2025.
2. C. Li, T. Ono, T. Uemori, **H. Mihara**, A. Gatto, H. Nagahara, Y. Moriuchi, "NeISF: Neural Incident Stokes Field for Geometry and Material Estimation," IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 21434–21445, 2024.
3. T. Sasaki, Y. Moriuchi, **H. Mihara**, K. Nakamura, J. Murayama, "Georeferenced Image Stitching with Pose Optimization Using Image Similarity with Little Overlap," Precision Agriculture '21, pp. 677–684, 2021.
4. Y. Moriuchi, K. Nakamura, **H. Mihara**, T. Sasaki, A. Ito, E. Itakura, H. Mori, J. Murayama, "Robust Vegetation Segmentation for Image-based Field Survey," Precision Agriculture '21, pp. 419–426, 2021.
5. J. Murayama, Y. Hirasawa, Y. Kondo, Y. Lu, T. Kurita, S. Kaizu, **H. Mihara**, T. Yamazaki, Y. Maruyama, Y. Uesaka, Y. Matoba, N. Terada, K. Komori, Y. Oba, S. Arakawa, K. Akiyama, Y. Oike, S. Sato, T. Ezaki, "Back-illuminated CMOS Image Sensor with Built-in Polarizer and Its Applications," JSAP Spring Meeting, 2018 (in Japanese).

6. **H. Mihara**, T. Funatomi, K. Tanaka, H. Kubo, H. Nagahara, Y. Mukaigawa, "4D Light Field Segmentation with Spatial and Angular Consistencies," IEEE International Conference on Computational Photography (ICCP), pp. 54–61, 2016.
7. **H. Mihara**, T. Iwaguchi, K. Tanaka, H. Kubo, H. Nagahara, Y. Mukaigawa, "Refocused Image Generation with Occluder Removal Using Light Field," IPSJ SIG-CVIM, 2015 (in Japanese).
8. **H. Mihara**, S. Asada, A. Ishihara, T. Iwaguchi, T. Okamoto, K. Tanaka, H. Kubo, Y. Mukaigawa, "Per-pixel Independent Refocusing Using Light Field," JCEEE Kansai, 2014 (in Japanese).

Patents

Granted

1. S. Kaizu, **H. Mihara**, T. Kamiya, S. Aotake, "Signal Processing Device, Signal Processing Method, and Distance-Measuring Module," US Patent 12,405,377, 2025.
2. **H. Mihara**, S. Kaizu, T. Kurita, Y. Hirasawa, "Information Generation Device, Information Generation Method and Program," US Patent 12,101,583, 2024.
3. **H. Mihara**, A. Shin, T. Narita, K. Hongo, M. Nakamura, "Control Device and Control Method for a Movable Body," US Patent 12,038,750, 2024.
4. **H. Mihara**, S. Kaizu, T. Kurita, Y. Kondo, Y. Hirasawa, "Image Processing Device and Image Processing Method," US Patent 11,663,831, 2023.
5. Y. Hirasawa, Y. Kondo, H. Oyaizu, S. Kaizu, T. Kurita, T. Zhuang, Y. Lu, **H. Mihara**, "Image Processing Device and Image Processing Method," US Patent 10,812,735, 2020.

Pending

1. **H. Mihara**, Y. Moriuchi, T. Uemori, L. Sun, "Image Processing Apparatus and Image Processing Method," US Patent App. 18/871,313, 2025.
2. M. Fujioka, Y. Moriuchi, K. Nakamura, T. Sasaki, **H. Mihara**, "Signal Processing Device and Signal Processing Method," US Patent App. 18/573,988, 2024.
3. M. Fujioka, Y. Moriuchi, K. Nakamura, T. Sasaki, **H. Mihara**, "Signal Processing Apparatus and Signal Processing Method," US Patent App. 18/580,008, 2024.
4. Y. Moriuchi, K. Nakamura, **H. Mihara**, T. Sasaki, "Image Processing Apparatus and Method," US Patent App. 17/918,500, 2023.
5. **H. Mihara**, S. Kaizu, "Signal Processing Device, Signal Processing Method, and Distance Measurement Device," US Patent App. 17/779,026, 2022.
6. **H. Mihara**, S. Kaizu, "Signal Processing Device, Signal Processing Method, and Ranging Module," US Patent App. 17/608,059, 2022.
7. S. Kaizu, **H. Mihara**, "Light Receiving Device and Method for Driving Light Receiving Device," US Patent App. 17/597,439, 2022.
8. H. Suzuki, A. Irie, T. Kasai, M. Nakamura, T. Narita, **H. Mihara**, "Control Device, Control Method, and Program," US Patent App. 17/250,328, 2021.

* *Several of the above are also granted/filed in JP, EP, CN, and DE as family members.*

Honors and Awards

Best Student Volunteer Award, SIGGRAPH Asia 2015 — Nov. 2015
SIGGRAPH Asia 2015, Kobe, Japan

Award, ITE Kansai Section — Apr. 2015
"Per-pixel Independent Refocusing Using Light Field," JCEEE Kansai

Grand Prize, NAIST CICP (Creative and International Competitiveness Project) — Mar. 2015
Occluder removal using a light field; selected as the best on-campus research project

Technology Demonstrations & Exhibitions

Presented in Person

Real-time Rendering Demo of NeISF and NeISF++ Jul. 2025
IEEE International Conference on Computational Photography (ICCP) 2025 Toronto, Canada

- Presented a real-time rendering demo of NeISF and NeISF++ (published at CVPR) at the Sony sponsor booth.

Real-time Rendering Demo of NeISF Jun. 2024
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2024 Seattle, USA

- Ran a live real-time rendering demo of NeISF (published at CVPR) for attendees at Sony's exhibition booth.

Real-time Demo of a Polarization Sensor and Its Applications Aug. 2022
IEEE International Conference on Computational Photography (ICCP) 2022 Pasadena, USA

- Demonstrated a polarization image sensor and its applications at the Sony sponsor booth, including a real-time system I built for polarization-based auto white balance (color constancy) based on research by my Sony colleagues (Ono et al., CVPR 2022).

Contributed (Research Collaboration)

Real-time Compositing of Live-action Footage and 3DCG via OpenTrackIO Jul. 2026
SIGGRAPH 2026 Los Angeles, USA

- Jointly developed a real-time compositing demonstration that combined the business unit's real-time production technology with our research technology (production-quality 3D geometry and material capture using NeISF); presented on-site by the collaborators to a computer-graphics research audience.

Real-time Compositing of Live-action Footage and 3DCG via OpenTrackIO Apr. 2026
NAB Show 2026 Las Vegas, USA

- Jointly developed a real-time compositing demonstration that combined the business unit's real-time production technology with our research technology (production-quality 3D geometry and material capture using NeISF); presented on-site by the collaborators to broadcast and media-production professionals.

Skills

Research Areas: Image Processing, Computer Vision, Computational Photography; sensor-oriented image processing (Polarization, ToF, Near-Infrared)

Machine Learning: Deep Learning, PyTorch (CNNs, differentiable rendering)

Data & Simulation: Synthetic dataset generation with Blender (CG); physically-based and differentiable rendering (Mitsuba); data-acquisition system design (sensor and illumination rigs)

Programming: Python, C++, CUDA, OpenCV

Tools: LaTeX, Microsoft Office, Arduino

Other Experience: Technical communication & demos, maintenance and upgrades of shared computing environments, organized and led a conference-participation report session for an audience of several hundred, contract/privacy compliance coordination with the legal department

Languages: Japanese (Native), English (Professional Working Proficiency)